COLOMBE EXAUCÉ

AI RESEARCH ENGINEER

INFORMATION

colombeexauce@gmail.com
linkedin.com/in/colombe-exauce/
https://github.com/cexauce

Driving License B

SKILLS

Deep Learning
Literature Review
Software Development
Data Analysis and Visualization
Signal/Image Processing
Documentation
Scientific writing
Scientific Communication

TOOLS

Python, C, C++, C#, Java
Matlab,

OpenCV, Pytorch, Matplotlib, Numpy,
Pandas, Numba, Anaconda, Jupyter,
Seaborn ScikitLearn, Numba

Visual Studio Code,
Git, Github, Doxygen, Jupyter
Linux, Windows

PERSONAL PROJECTS

- 3D Platform game with Unity in C#
- Matrix Library in C++
- Script for parallel downloading video/audio using multi-threading

LANGUAGES

French : mother tongue

English : professional/fluent
Spanish : beginner
Swedish : beginner

HOBBIES

Music (guitar, bass, piano)
Gym
Bouldering

ACADEMIA

PUBLICATION : 1st Author ; Interspeech 2026 : Audio-visual Contrastive Alignment for Diffusion-based Visual-conditioned Speech Enhancement

SUMMER SCHOOL 2025 : EURASIP Machine Learning and Signal Processing

PROFESSIONAL EXPERIENCE

AI RESEARCH ENGINEER

INRIA Lorraine, Nancy | Feb. 2025- Dec. 2026

Audio-Visual Fusion for Speech Enhancement (AVSE)

- Scientific Literature review (Multimodal Representation Learning and AVSE)
- Developing novel methods for Audio-visual Fusion in AVSE
- Training, evaluating and inspecting audio-visual contrastive models and diffusion models
- Presenting research work
- Writing, publishing and presenting at INTERSPEECH 2026

AI RESEARCH INTERN

Mines ParisTech | Apr.-Sep. 2024

Unsupervised Learning Driving Interactions for Autonomous Driving

- Scientific literature review
- Processing GPS data, extracting interactions for training
- Adapting the MoCo algorithm for driving interactions
- Inspecting & visualizing results

SOFTWARE & COMPUTER VISION ENGINEER

MAF RODA, Montauban | Jul. - Dec. 2023

- image processing : fruit segmentation on multi-spectral images
- software : improving an HMI to control an armed robot

MACHINE LEARNING INTERN

HiFive-ARISE | Nov. 2022 - Jan. 2023

Anomaly Detection, Data analysis and visualization

AI RESEARCH INTERN

University of Lund (Sweden) | Nov. 2021 - May 2022
Deep Reinforcement Learning with a focus mechanism

- Scientific Literature Review on DQN
- Implementing and training a state-of-the-art RL algorithm
- Evaluating, comparing results and writing

C++ & IMAGE PROCESSING ENGINEER

ODESYO in Montpellier | Mar 2019 - Feb 2020

EMBEDDED SOFTWARE ENGINEER

ESSILOR-CNRS in Toulouse | Apr. 2017 - Jun 2018

IMAGE PROCESSING INTERN

Valeo in IdF | May - Sep. 2016

EDUCATION

MOOC Coursera Deep Learning Specialisation | 2023

Master's Degree in Data Science, AI and Machine Learning
Halmstad University, Sweden | 2020-2022

Master's in Electronics & Telecommunications
ENSIL-ENSCI in Limoges, France | 2013 - 2016

PREPARATORY CLASSES: MPSI-PSI
lycée Descartes-Vaucanson in Tours, France | 2010 - 2013